

MECHANICAL DYNAMICS OF CONTROL HANDOVER: STEP DISCONTINUITY VS. C¹ BUMPLESS BLENDING

Eliminating Mechanical Jerk Shock, Drive Slip, and Integrator Windup via Dynamic Blending $\tau_{blend} \geq 1.5|\Delta u| / j_{allowable}$

× UNBLENDED STEP HANDOVER (Naïve Hard Switch, $\tau_{blend} = 0$ ms)

$a(t)$

Weight



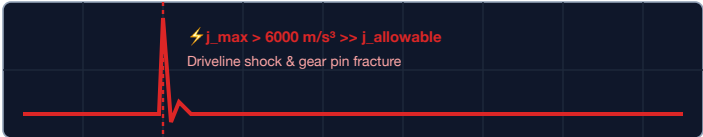
$u(t)$

Torque



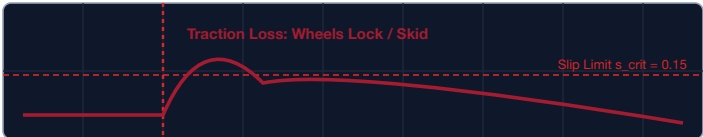
$j(t)$

Jerk



$s(t)$

Tire Slip

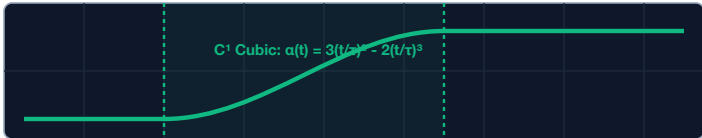


FAILURE: Infinite Jerk · Mechanical Gearbox Shock · Stopping Path +3.67 m

✓ C¹ CUBIC BUMPLESS BLEND (Pre-Biased, $\tau_{blend} = 750$ ms)

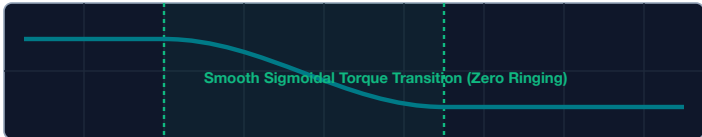
$a(t)$

Weight



$u(t)$

Torque



$j(t)$

Jerk



$s(t)$

Tire Slip



SUCCESS: Zero Shock · Continuous Acceleration · Integrators Pre-Biased to $x(t)$